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Quantitative haplotype-resolved analysis of mitochondrial DNA heteroplasmy in Human single oocytes, blastoids, and pluripotent stem cells

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ABSTRACT

Maternal mitochondria are the sole source of mtDNA for every cell of the offspring. Heteroplasmic mtDNA mutations inherited from the oocyte are a common cause of metabolic diseases and associated with late-onset diseases. However, the origin and dynamics of mtDNA heteroplasmy remain unclear. We used our individual Mitochondrial Genome sequencing (iMiGseq) technology to study mtDNA heterogeneity, quantitate single nucleotide variants (SNVs) and large structural variants (SVs), track heteroplasmy dynamics, and analyze genetic linkage between variants at the individual mtDNA molecule level in single oocytes and human blastoids. Our study presented the first single-mtDNA analysis of the comprehensive heteroplasmy landscape in single human oocytes. Unappreciated levels of rare heteroplasmic variants well below the detection limit of conventional methods were identified in healthy human oocytes, of which many are reported to be deleterious and associated with mitochondrial disease and cancer. Quantitative genetic linkage analysis revealed dramatic shifts of variant frequency and clonal expansions of large SVs during oogenesis in single-donor oocytes. iMiGseq of a single human blastoid suggested stable heteroplasmy levels during early lineage differentiation of naïve pluripotent stem cells. Therefore, our data provided new insights of mtDNA genetics and laid a foundation for understanding mtDNA heteroplasmy at early stages of life.

INTRODUCTION

Mitochondria and their 16.5-kb circular genomes (mtDNA) undergo constant turnover in arrested primary oocytes for decades in humans (1,2). Frequent DNA replication and lack of efficient DNA repair mechanism in the mitochondrion contribute to a mutation rate that is ∼10-fold higher than that of the nuclear genome (2–4). These features of mitochondria biology mean that the hundreds of thousands of copies of mtDNA in the oocyte could exist as an admixture of wild-type and mutant mtDNA (termed heteroplasmy) (5) and these mtDNA will be the sole source of mitochondria for every cell of the

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offspring. Sequencing studies of human populations show that germline heteroplasmy affects nearly fifty percent of humans (6,7). Heteroplasmic mutations inherited from the oocyte, if present above a heteroplasmy threshold, are among the most common causes of inherited metabolic diseases (8,9), and are also associated with late-onset diseases (3).

The origin of pathogenic mtDNA variants remains unclear. Heteroplasmic mutations have been documented in healthy human oocytes and primordial germ cells (10,11). Experimental data and population genetics modeling suggest that heteroplasmies arisen in mature oocytes strongly influence the inheritance of mtDNA mutations in the offspring (1,2). Clonal expansion of mtDNA mutations that are generated very early in life has been suggested to drive mitochondrial dysfunction in aged human colorectal epithelium (12). Yet, many key questions remain unanswered. For example, are the ostensibly somatic mutations found in old individuals originated from low-level heteroplasmic mutations in the oocyte or arose de novo? What is the level of mtDNA genetic heterogeneity among oocytes of the same female after the mtDNA genetic bottleneck? What is the functional significance of genetic linkage between variants? How to track the dynamics of maternally inherited heteroplasmy in early embryonic development? Answering these questions is challenging because there is a lack of quantitative method for the analysis of the multitude of mitochondrial genomes at the individual-molecule level in single cells, which would be required to study the ontogeny, heterogeneity, dynamics, and genotype-phenotype relationship of mtDNA mutations.

Most mtDNA sequencing methods are based on short-read shotgun or amplicon next-generation sequencing (NGS) (1,13–17). Also, many conclusions are based on imputation of data from bulk sequencing in somatic cells rather than actual measurements in single oocytes. Bulk sequencing masks variants in rare cells and underestimates cell-to-cell heterogeneity. Although sequencing mtDNA amplicons from single cells is possible (18), heteroplasmy inferred from PCR products amplified from a heterogeneous population of mtDNA can be inaccurate due to PCR bias (19). Short-read NGS struggles with large structural variant (SV) detection and false variants due to nuclear mitochondrial DNA-like sequences (NUMTs) (2,14,15) and cannot provide true haplotype (or variant linkage information) of individual mtDNA.

To overcome these challenges and gain new insights into mtDNA genetics in the female germline, we used our recently developed long-read sequencing strategy–individual Mitochondrial Genome sequencing (iMiGseq, see the accompanying manuscript) – to perform high-throughput analysis of the whole landscape of mtDNA heteroplasmy in single oocytes and blastoids. iMiGseq uses a simple and fast procedure to label individual mtDNA in lysed single cells without labor-intensive mitochondria isolation steps (18,20). This allows base-resolution analysis of up to tens of thousands of full-length mtDNA per cell. Our study uncovered whole spectrum of pathogenic mtDNA mutations, including SNVs and large SVs, in single cells that lie below the current 1% detection limit (2,14,15). It illustrated sequential acquisition of detrimental mutations in defective mtDNA in human oocytes. It revealed dramatic shifts in variant frequency and clonal expansion of SVs during oogenesis in humans and mice and stable heteroplasmy levels during human blastoid generation. Our study showed the first haplotype-resolved mitochondrial genomes from single human oocytes and single blastoids. It revealed the linkage of rare heteroplasmic mutations, illustrated the genetic linkage between pathogenic mtDNA mutations, and could facilitate the understanding of the complexity of mitochondrial diseases.

MATERIALS AND METHODS

Cell lines and human oocytes

All human immature oocytes were collected from the reproductive medical center in the Third Affiliated Hospital of Guangzhou Medical University. The study of human oocytes collection was approved by the Institutional Review Board (IRB) of the Third Affiliated Hospital of Guangzhou Medical University and KAUST Institutional Biosafety and Bioethics Committee (IBEC). All of the oocyte donors signed the inform consent voluntarily after they were clearly informed all of the content and details of the experiments. The women were in intracytoplasmic sperm injection (ICSI) cycles because of male infertility were involved in this study, and the immature oocytes were found after oocyte retrieval and cumulus cells removal. In general, such immature oocytes will be discarded as a medical waste and not used for the fertilization during the process of assisted reproductive technology. The endometrium tissues were donated by women with mitochondria diseases who signed the informed written consent voluntarily after learning the study aims and methods. The protocol was approved by the Institutional Review Board of Peking University Third Hospital. Endometrium tissues were collected into a 50 ml centrifuge tube with DMEM/F12 culture media (no phenol red, Invitrogen, Carlsbad, CA, USA) containing 100 U/ml penicillin and 100 μg/ml streptomycin (Invitrogen) during a routine gynecological examination. The tube with endometrial sample was put in the ice, and quickly transported to the laboratory. The endometrial tissue was rinsed using Hanks buffered salt solution (HBSS, Invitrogen) three times and minced into 1 mm^3 fragments. After digested with 2 mg/ml collagenase type I (Life Technologies, New York, NY, USA) for 45 min and DNase I for 30 min subsequently, the dissociated cellular suspension was treated with ACK lysis buffer (Life Technologies, New York, NY, USA) to remove red blood cells. The cells were then centrifuged at 200 g for 10 min and resuspended using DMEM/F12 supplemented with 10% fetal bovine serum (Charcoal/Dextran Stripped, Gemini, California, USA), and plated on 35 mm dishes (Corning) at 37°C in 5% CO_2.

In vitro maturation of human oocytes

Human immature oocytes were cultured following our previous protocols which was consist of IVM basal medium, 0.075 IU/ml FSH, 0.075 IU/ml luteinizing hormone (LH), 10 ng/ml EGF, 10 ng/ml BDNF and 10 ng/ml IGF-1 (21). The oocytes expelling the first polar body were regarded as mature oocytes at metaphase II stage. Single matured MII

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oocyte was transferred into 0.5ml EP tube without medium using mouth pipette and frozen in –80°C refrigerator before further molecular analysis.

Cell preparation and UMI labeling

Single oocytes were obtained by mouth pipette and transferred into cold PBS under a stereo microscope. One microliter of single oocytes were lysed in 5 μl RIPA buffer (150 mM sodium chloride, 1.0% NP-40, 0.5% sodium deoxycholate, 0.1% sodium dodecyl sulfate, 50 mM Tris, pH 8.0) on ice for 15 min. The cell lysate was diluted by adding 10 μl H_2O, and further purified to extract total DNA using 15 μl Beckman Coulter AMPure XP beads (A63882). Two-round of 70% ethanol washes were performed to remove detergents. The DNA was eluted in 10 μl H_2O to be used for the UMI labeling of mtDNA followed the previous protocol (see the iMiGseq paper). The full list of oligos used in this study is shown in Supplementary Table S1.

Human naïve iPSC reversion and blastoid generation

We generated niPSCs and human blastoids as previously described with minor modifications (22,23). Briefly, primed iPSCs were maintained on an irradiated MEF feeder with KSR/FGF medium, then, they were exposed for 3 days to N2B27 supplemented with 1 uM PD0325901, 1 mM valproic acid sodium salt, and 10 ng/ml human LIF, and then propagated in N2B27 medium supplemented with 1 uM PD0325901, 2 uM XAV939, 2 uM Goö 6983 and 10 ng/ml human LIF. After 10 days, domed-shaped niPSCs appeared in culture and were passaged routinely using accutase every 3–4 days. To make human blastoids, niPSCs were dissociated into single cells, gelatinized for 60 min, and floating cells were collected and counted. Cells were washed with DMEM/F12 with 0.1% BSA, and resuspended in N2B27 supplemented with 1.5 uM PD0325901, 1 uM A83-01 and 10 uM Y-27632 seeded into AggreWell 400 wells at a density of 20–30 cells per microwell. The medium was switched after 2 days to N2B27 supplemented with only 0.5 μM A83-01 and maintained to day 5–6, and further processed for downstream analysis. All cell culture was maintained in an incubator with 5% CO_2 and 5% O_2. The starting population of niPSCs were manually picked as colonies and transferred to individual PCR tubes (approx. 500 cell/tube). One tube was used for iMiGseq. Single Day-5 blastoids (cell number: 184 ± 49/blastoid) were examined under a Leica M205 FCA stereo microscope, and single blastoids that contained a cavity, an ICM-like cell cluster, and a well-formed trophectoderm layer were manually picked and transferred to individual PCR tubes. Each tube was visually inspected under a Leica M205 FCA stereo microscope to ensure that only one blastoid was present. One tube was used for iMiGseq.

Sequencing and bioinformatic analysis

All samples were sequenced in the Oxford Nanopore Technologies MinION using one R9.4.1 flow cell per sample. Library preparations were done using the ligation sequencing kit (Oxford Nanopore Technologies, SQK-LSK109). Base calling was done using Guppy (v3.2.1 for oocytes and v5.0.7 for blastoids). All procedures were preformed according to manufacturer's protocols. Bioinformatic analysis was performed using VAULT following the previous protocol (see the iMiGseq paper).

The synonymous and nonsynonymous sites of mouse mitochondrial genome was calculated using the *pS_pN_count* tool of VAULT, and based on the Nei-Gojobori method (24). The dN/dS ratio was calculated using bash command based on the Nei-Gojobori method. The mutational spectrum was analyzed using SomaticSignatures (25).

RESULTS

Whole landscape of mtDNA heteroplasmy in single human oocytes

We have developed a quantitative and sensitive method–iMiGseq–that could provide an unbiased analysis of all types of genetic variants in mtDNA. We hypothesize that iMiGseq could improve our understanding of mtDNA heteroplasmy in human zygotes. As a proof of concept, we applied iMiGseq to six human oocytes–three from three independent donors (referred to as hOOs hereafter) and the other three from a single female (referred to as SFhOOs)–that were matured to the metaphase II stage *in vitro* (see Methods). iMiGseq of human oocytes captured thousands of extensive-coverage (covering > 95% of the full-length mtDNA with depth ≥3×) mtDNA (Table 1, Supplementary Figure S1a, b). Variant analysis identified a significant amount of SNVs in all samples and the vast majority of them had a nominal heteroplasmy level (NHL, similar to VAF, calculated by [no. of UMI groups with SNV]/[no. of UMI groups covering the SNV position]) below the 1% detection limit of heteroplasmy reported in published studies (2,13–15) (Figure 1A–D, Supplementary Figures S2a and S3a). The validity of rare variants detected by iMiGseq has been validated by PacBio sequencing, Sanger sequencing, Illumina NGS, digital droplet PCR (ddPCR), mutation spectrum analysis, and in silico simulations (see the iMiGseq paper). To avoid the influence of potential homoplasmic variants in the study of genetic heterogeneity of mtDNA, we excluded SNVs with NHL above 60% from most analyses detailed below. It is worth noting that NHL of the rarest SNVs (e.g. appear in one UMI group) tended to be more affected by the sequencing depth and may show a higher variability.

The high-frequency SNVs (NHL ≥ 1%) in hOOs belonged to the mtDNA haplogroups M and F, which is consistent with east Asian origin of the donors (26,27) (Figure 1E, F). Interestingly, the low-level heteroplasmic SNVs of the three independent donors were markedly more diverse (Figure 1E). Note that haplogroup typing by iMiGseq is done in the context of full-length individual mtDNA in one assay as opposed to multiple genotyping assays as used currently.

As seen in hOOs, most SNVs discovered by iMiGseq in SFhOOs had an NHL <1% (Figure 1B, Supplementary Figure S3a). The three SFhOOs shared a large proportion (93.5%) of high-frequency SNVs, consistent with their single donor origin (Figure 1G). However, an unappreciated level of heterogeneity was observed in the three SFhOOs when low-frequency SNVs (NHL < 1%) were

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Table 1. Summary of individual sequencing runs

Sample	Mapped read count	Reads with UMI	Detected mtDNA	mtDNA with SNV	SNV count	Median covered region of UMI groups	Extensive coverage mtDNA*	Unique SNV number	Somatic SNV (<0.01 VAF)*	Somatic SNV load per mb*
hOO_1	763 582	95 387	2308	2301	75 483	16 531	2072	141	128	3.7
hOO_2	397 426	64 646	1461	1457	32 276	16 413	1009	81	59	3.6
hOO_3	506 082	99 404	1535	1514	22 522	15 568	712	69	20	1.7
SFhOO_1	419 712	143 679	2866	2859	96 048	16 543	2548	266	168	4.0
SFhOO_2	189 803	100 900	1048	1047	37 608	16 560	923	224	118	7.7
SFhOO_3	362 527	113 042	2085	1883	55 009	16 489	1599	234	76	2.9
niPSCs	586 901	120 057	245	238	2201	10 438	104	44	1	0.59#
blastoid	463 472	156 767	632	622	8525	16 253	440	76	65	9.03

*UMI groups with reads covering > 95% of the full-length mtDNA with depth ≥3×.
#Somatic SNV load estimate may be inaccurate due to low number of extensive-coverage mtDNA.

compared (Figure 1G). Given that the six previously sequenced mouse oocytes also shared a very small fraction of the SNVs despite being from the same female (Figure 1H, see the iMiGseq paper), our observations were consistent with an independent germline bottleneck in the oocyte lineage (1), and revealed a surprisingly large genetic heterogeneity among oocytes of the same female in humans. Such genetic heterogeneity in mtDNA has been underestimated by conventional mtDNA sequencing.

We found that, among shared SNVs in SFhOOs, those with high NHLs had stable heteroplasmy levels among the oocytes (average correlation coefficient 0.94), while low-NHL SNVs were much more variable (average correlation coefficient 0.80). One SNV (m.309C > T) showed low NHL (0.4% in SFhOO_1 and 0.8% in SFhOO_3) in two oocytes but a much higher NHL in the third oocyte (3.3% in SFhOO_2). Similarly, we also discovered several heteroplasmic variants that showed dramatically different frequencies in different oocytes of the same female mouse (see Figure 2E of the iMiGseq paper). These results suggested that rare variants could under some circumstances accumulate during oogenesis, leading to dramatic shifts in variant frequency. Our data thus provided a sensitive and quantitative method for studying mtDNA segregation in oocytes and suggested that low-frequency variants are subjected to a much higher level of heteroplasmy fluctuation, some of which may be mistaken as absent by ensemble sequencing.

We assembled consensus mtDNA sequences using reads from extensive-coverage UMI groups and obtained 125, 81 and 46 full-length mtDNA for hOO_1, hOO_2 and hOO_3, respectively. MITOMASTER (26) assigned all assembled mtDNA of hOO_2 and hOO_3 to the M7c1c2 and F1a1a haplogroups, respectively. For hOO_1, 122 assembled mtDNA were assigned to M7b1a1b, while the remaining three contained a different allele (m.3483G > C versus A) consistent with the M7b1a1 haplogroup. Importantly, all assigned haplogroups matched the ethnic origin of the donors (Figure 1F). These data provided base-resolution haplotypes of individual mtDNA in single cells.

Previous reports based on deep NGS sequencing of human mtDNA identified several conserved replication related regions (e.g. MT-LSP) that were devoid of heteroplasmy (1% detection limit) (6,7). Interestingly, iMiGseq identified two SNVs in such purported 'variant-deserts' in two of four human samples with NHLs of 0.24% (m.438C > A in 293T cells) and 0.09% (m.16427C > T in hOO_2). Thus, using the unbiased ultra-sensitive iMiGseq, our data suggested that mtDNA mutations can randomly happen in any region, and that the heteroplasmies in 'variant deserts' are probably very rare and missed by conventional NGS, which is consistent with the view that mutations in these regions impair mtDNA propagation (6,7).

The Cancer Mitochondrial Atlas (TCMA (15)) provided a comprehensive collection of heteroplasmic mutations in human cancers. However, the ontogeny of these cancer-associated mtDNA mutations whose occurrence positively correlates with age (15) remains unclear. Interestingly, each of the human oocytes shared a significant portion of its unique SNVs with the TCMA database (15) (Figure 2A, Supplementary Figure 3b). We refined the comparison by separating unique SNVs according to their NHL (<1%), which showed that a significant amount of shared SNVs existed well below the 1% detection limit of most published studies (Supplementary Figures S2b and S3c). Further intersecting SNVs in human oocytes with TCMA-hotspot SNVs (SNVs exist in more than four samples/cancer types in TCMA database) showed that 9% of TCMA-hotspot SNVs were found in the six human oocytes sequenced by iMiGseq (Supplementary Figure S4). Although the biological significance of these cancer-associated mutations is unclear, these data suggested that some of them could be inherited through the germline of healthy women.

Additionally, intersecting iMiGseq SNVs with confirmed pathogenic mtDNA mutations (26) identified two SNVs causing Leber's hereditary optic neuropathy (LHON) and mitochondrial encephalopathy, lactic acidosis, and stroke-like episodes (MELAS, 3376G > A/p.Glu24Lys in *ND1*), and chronic progressive external ophthalmoplegia (CPEO, 5690A > G in $tRNA^{Asn}$), respectively (Figure 2A and B, Supplementary Figure S2b and c). Each was in one mtDNA in hOO_2 with an NHL of 0.07%. These data provided evidence that low-level disease-associated heteroplasmic mutations commonly exist in single healthy human oocytes.

Mutational spectrum analysis showed the high-level heteroplasmic SNVs (1–60% NHL) in all six human oocytes were predominantly C > T and T > C (Supplementary Figure S5), consistent with our iMiGseq data of 293T cells (see Supplementary Figure S3 of the iMiGseq paper) and with

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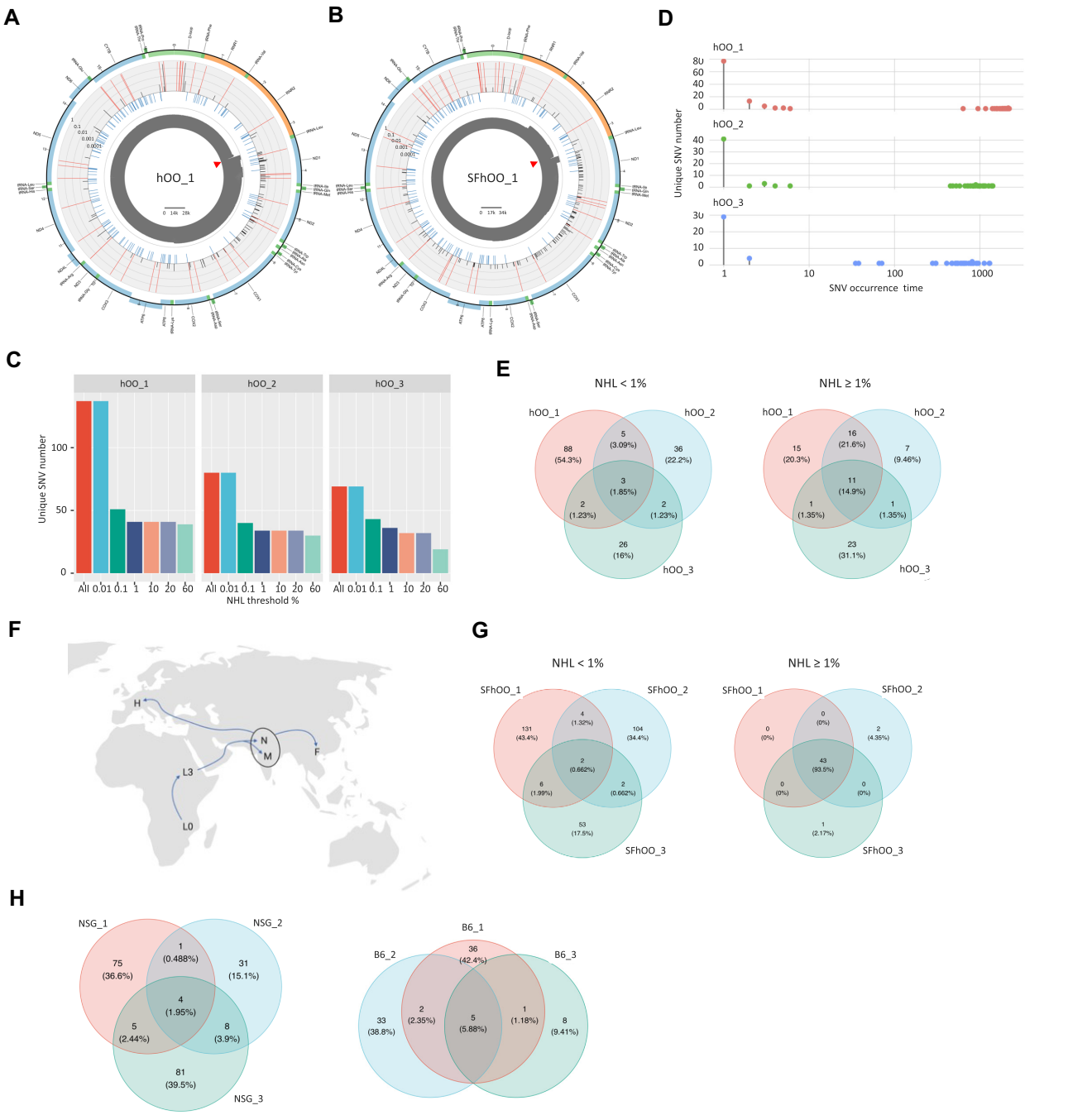


Figure 1. Unbiased analysis of the whole mutational landscape of mtDNA in single human oocytes. (**A**) Circular plots showing the distribution of SNVs in the mitochondrial genome of hOO_1. The innermost circle (grey) shows the depth of reads of all detected UMI groups in linear scale as indicated by the scale bar in the center. The red triangle indicates the position of the primers. The middle circle (light blue) represents common SNPs from the human dbSNP-151 database. Individual SNVs are plotted in the outer barplot circle, in which the height of bar represents the VAF. Red color indicates VAF >0.6. The outermost circle is a color-coded diagram of the human mtDNA. Blue: protein-coding genes. Yellow: rRNA genes. Dark green: tRNA genes. Light green: D-loop. (**B**) Circular plots showing the distribution of SNVs in the mitochondrial genome of SFhOO_1. The arrangement of the circular plots is similar to (A). (**C**) Unique SNV number above different heteroplasmy threshold. For all three hOOs, a significant amount of unique SNVs had an NHL below the 1% detection limit of current methods. (**D**) Occurrence time of unique SNVs detected in iMiGseq of human oocytes. Most of the unique SNVs appear one time (exist in one UMI group). (**E**) Venn diagrams showing the overlap of mtDNA SNVs between the hOOs at different heteroplasmy levels. A higher proportion of SNVs with an NHL ≥1% (right) were shared among hOOs than those with an NHL <1% (left). (**F**) A map showing the mtDNA phylogeny and geographic distribution of the haplogroups of the three hOOs in this study. The 11 SNVs shared by all three hOOs separate the L3 haplogroup from its ancestral L0 haplogroup. These SNVs are retained in Asian haplogroups M and F (a daughter group of N) but not in the European haplogroup H, which is consistent with the east Asian origin of the donors. (**G**) Similar to (**E**) but for SFhOOs. (**H**) Venn diagrams showing the overlap of mtDNA SNVs detected in the three NSG (left) and B6 (right) mouse oocytes.

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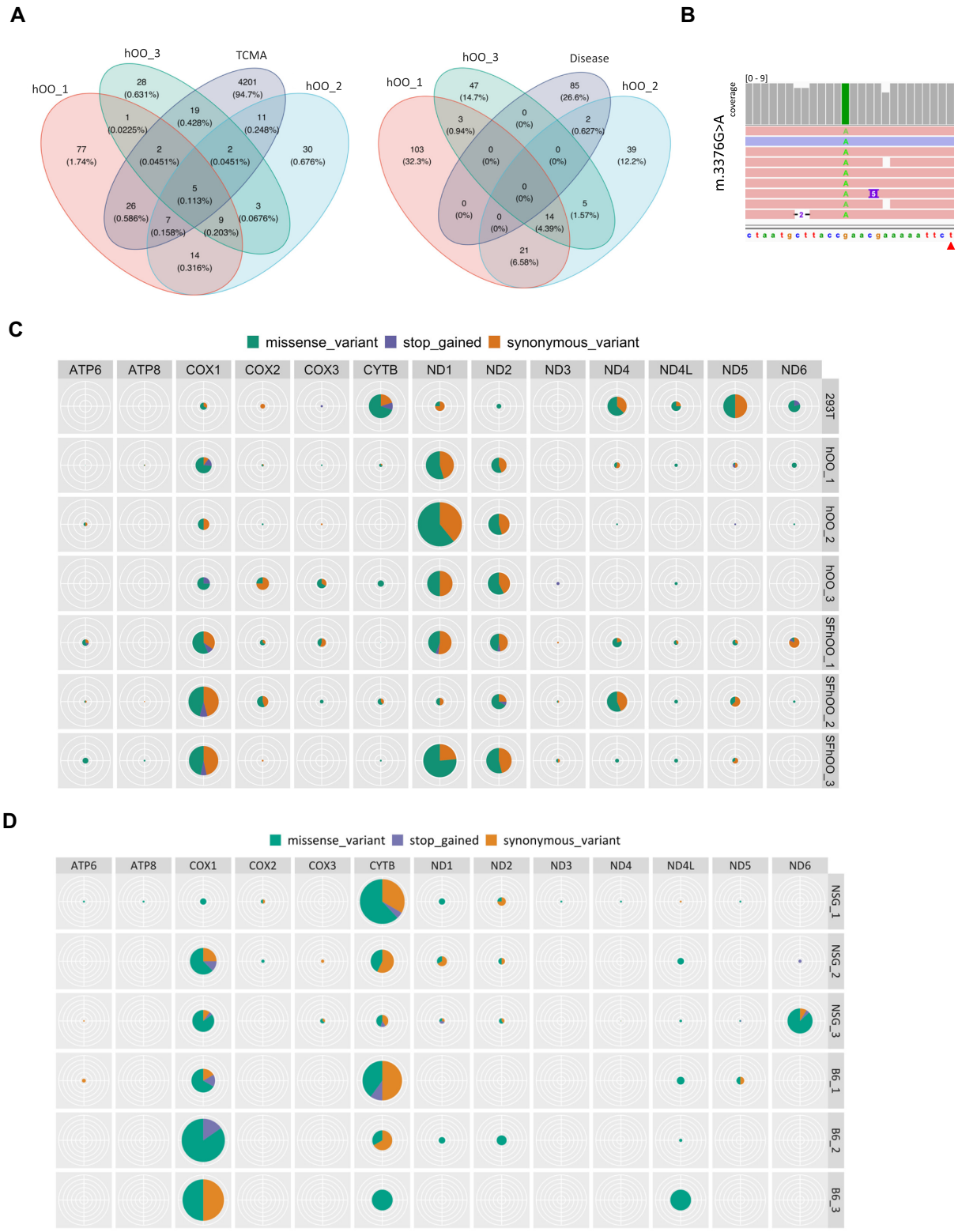


Figure 2. Functional analysis of mtDNA variants in single human and mouse oocytes. (**A**) Venn diagrams showing the overlap of all mtDNA SNVs in hOOs and cancers (TCMA, left), or confirmed pathogenic mtDNA SNVs in mitochondrial diseases (Disease, right). Cancer-related mtDNA SNVs exist in all three hOOs. (**B**) The disease-related SNV detected in hOO_2 shown in IGV tracks. The mutant bases are indicated by red triangles under the plots. (**C**) Functional annotation of SNVs in protein-coding regions detected by iMiGseq of hOOs, SFhOOs, and the 293T cell line (sequenced by Nanopore). Variants called in all UMI groups (not limited to extensive-coverage groups) were used to generate this plot. The size of the pie chart is proportional to overall mutation frequency, and the color corresponds to the type of mutation. All human oocytes show a similar enrichment of SNVs in *COX1*, *ND1* and *ND2* genes. The 293T cell line has more SNVs in the *CYTB*, *ND4* and *ND5* genes. (**D**) similar to (**C**) but for single mouse oocytes.

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previous data (28). The mutational spectrum of rare variants (NHL < 1%) of full-length mtDNA was previously inaccessible due to technical limits. iMiGseq showed for the first time, a diverse spectrum of rare variants (NHL < 1%) in single human oocytes (Supplementary Figure S5). Lastly, data from iMiGseq showed that oxidative damage (causing C > A transversions in either H or L strands) is not a major source of de novo mtDNA mutations in the human and mouse germline (Supplementary Figure S6). A similar observation was also made in somatic tissues (17).

Mutation annotation illustrated that variants in rRNA genes were significantly depleted in five of the six human oocytes (the sample number of hOO_3 was too small for proper statistical testing, (Tables 2 and 3). Among protein-coding genes, all human oocytes consistently showed frequent mutations in the *ND1*, *ND2* and *COX1* genes, which is distinct from the pattern of 293T cells (Figure 2C). These cell-type specific mutation patterns further argue against the possibility that iMiGseq variants are artifacts introduced during library preparation, because such artifactual mutations would be the same in all cell types. Deleterious SNVs (missense and stop-gained variants) were prevalent among heteroplasmic SNVs in all human oocytes (Figure 2C, Supplementary Figure S7), an observation that also holds true in all mouse oocytes (Figure 2D). Synonymous mutations occurred significantly more frequently than expected by chance, while non-synonymous mutations did not deviate from random chance (Tables 2 and 3).

Genetic linkage analysis reveals sequential acquisition of mutations and clonal expansion of large SVs in human oocytes

The functional significance of an mtDNA variant is often strongly modified by other co-inherited variants (3). One of the advantages of iMiGseq is its ability to provide base-resolution full haplotype, which greatly facilitates the study of linkage between heteroplasmy variants. Several phylogenetic trees were constructed based on haplotypes of individual mtDNA in human oocytes, which showed, for the first time, the evidence of sequential acquisition of *de novo* heteroplasmic SNVs in individual mtDNA (Figure 3A). The variant linkage information from complete and haplotype-resolved mtDNA sequences was only possible due to the analysis of individual mtDNA in single oocyte and would have been missed by conventional methods.

Large SVs ranging from 2580 to 7733 bp were detected in three SFhOOs (three deletions in SFhOO_1, twelve deletions in SFhOO_2, and one deletion in SFhOO_3). No sequence similarity or repeat elements were found around the breakpoint junctions of any deletion, suggesting a random acquisition of large variants. Significantly, we observed cases of potential clonal expansion of mutant mtDNA harboring the same large deletion in SFhOO_2 (Figure 3B). The SV-containing mtDNA shared the same polymorphisms as wild-type mtDNA from the same oocyte, thus ruling out the possibility of NUMT contamination. No common SVs were identified among the three SFhOOs, suggesting independent mutagenesis events. SFhOO_2 showed a significantly higher SV load with 0.95% of mtDNA harboring large deletions, as compared with 0.07% in SFhOO_1 and 0.05% in SFhOO_3. Our data suggested health human oocytes frequently harbor heterogeneous SVs, some of which can be amplified during mtDNA replication. By revealing their existence iMiGseq enables the tracking and analysis of these SVs in functional studies in the future.

Evidence of purifying selection in rare variants

The dN/dS ratio is a major tool used to detect selection on protein-coding mutations (29), and has been used to study selection on mtDNA (15,30–32). The dN/dS ratios of somatic SNVs (NHL < 60%) of all mtDNA in each of the six human oocytes showed strong evidence for purifying (negative) selection against non-synonymous mutations, which was also reported in a previous study (12) (Figure 3C, Supplementary Figure S3d). Considering that oocytes experience an mtDNA bottleneck during germ cell development, our data provided single-mtDNA level evidence that supports the role of the germline bottleneck in eliminating deleterious mutations in the offspring.

Tracking mtDNA heteroplasmy dynamics during the generation of human blastoid from naïve pluripotent stem cells

The dynamics of the heteroplasmic mtDNA variants present in a zygote during human development is poorly explored due to technological and ethical limitations. We and others recently developed human pluripotent stem cell (PSC)-based blastocyst-like structures termed blastoids to model early development and embryogenesis (23,33–37). We therefore next tested if iMiGseq could be used to track heteroplasmy dynamics in the blastoid model of human early development (Figure 4A).

Human naïve PSCs (nPSCs) represent an early stage of pluripotency that differs from the primed pluripotency state in morphology and gene expression, and more importantly, by retaining the potential to give rise to extra-embryonic tissues (38) (Figure 4B, C). In this study, we generated human blastoids from human naive iPSCs (niPSC) (see Materials and Methods) (Figure 4D). Days 5/6 human blastoids resembled natural blastocysts in size (~200 um) and in 3D organization that consisted of a cavity, an ICM-like cell cluster, and an encircling trophectoderm-like layer (Figure 4E, F). We applied iMiGseq to human niPSCs and niPSC-derived single blastoid (Table 1). Variant analysis found no SV in either niPSCs or blastoid, and the levels of heteroplasmic variants remain stable during blastoid generation (Figure 4G, H). Functional analysis of rare variants identified the emergence of 14 amino acid-changing mutations in the blastoid such as m.6249G > A (p.Ala116Thr) detected in 4 UMI groups (VAF of 0.71%) and annotated as potentially pathogenic by MITOMAP; and m.12700C > T (p.Leu122Phe) detected in 4 UMI groups (VAF of 0.8%) and not reported in MITOMAP (Figure 4I). The long-term influence of these rare mutations on blastoid development remains to be elucidated in the future and the sensitive iMiGseq method can help to track and study the dynamic changes of rare mtDNA mutations.

Despite the low sequencing depth, our data suggested that *in vitro* blastoid generation is less likely to affect the overall mtDNA heteroplasmic levels and that a sensitive method is needed to monitor the potential pathogenic mu-

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Table 2. The distribution of SNVs (NHL < 0.01) among mtDNA regions in hOOs

			hOO_1			hOO_2			hOO_3		
Region	bp	Expected %	Expected	Observed	*P*-value‡	Expected	Observed	*P*-value‡	Expected	Observed	*P*-value‡
D-loop	1122	6.8%	8.9	10	0.84*	4.0	6	0.44*	2.4	2	0.94*
tRNA	1508	9.1%	11.9	10	0.67	5.4	3	0.40*	3.2	2	0.69*
rRNA	2513	15.2%	19.9	5	**0.00 045**	9.0	3	**0.047***	5.3	2	0.18*
Synonymous	2834	17.1%	22.4	35	**0.005**	10.1	19	**0.0036**	6.0	11	**0.043***
Non-synonymous	8533	51.5%	67.5	71	0.60	30.4	28	0.62	18.0	18	0.87
Total	16569#			131†			59†			35†	

#Due to overlapping annotation the total number is different from the sum of all categories.
†Include low-coverage groups.
‡Two-tailed binomial test.
*Data not met the requirements for a *z*-test/sample number is smaller than 10.

Table 3. The distribution of SNVs (NHL < 0.01) among mtDNA regions in SFhOOs

			SFhOO_1			SFhOO_2			SFhOO_3		
Region	bp	Expected %	Expected	Observed	*P*-value‡	Expected	Observed	*P*-value‡	Expected	Observed	*P*-value‡
D-loop	1122	6.8%	13.4	16	0.55	9.0	6	0.39*	5.6	14	**0.0006***
tRNA	1508	9.1%	17.9	41	**<0.000 001**	12.0	24	**0.0005**	7.6	7	0.98*
rRNA	2513	15.2%	29.9	9	**0.00 005**	20.1	5	**0.0004**	12.6	0	**0.0002**
Synonymous	2834	17.1%	33.7	56	**0.000 036**	22.6	38	**0.0005**	14.2	21	0.06
Non-synonymous	8533	51.5%	101.5	69	**0.000 006**	68.0	55	0.03	42.7	39	0.48
Total	16569#			197†			132†			83†	

#Due to overlapping annotation the total number is different from the sum of all categories.
†Include low-coverage groups.
‡Two-tailed binomial test.
*Data not met the requirements for a *z*-test/sample number is smaller than 10.

tations. We envision that iMiGseq and human blastoid together will provide new opportunities for modeling mtDNA dynamics in both developmental biology and mitochondrial disease settings. Additionally, the ability of iMiGseq to analyze all types of variants in thousands of full-length mtDNA in single (or a small number of) cells may improve the clinical practice of preimplantation genetic diagnosis (PGD) of mitochondrial disease, and our data show that human blastoid offers a good system to model this.

DISCUSSION

Here, we presented an unbiased high-throughput base-resolution analysis of individual full-length mtDNA in single oocytes. By using a method that accurately detect heteroplasmy well below the detection limit of conventional NGS, we showed that most mtDNA SNVs in oocytes are below 1% VAF. They suggest that a hitherto underestimated population of rare de novo mtDNA variants exist in the female germline.

The analysis of oocytes from the same female showed that the frequency of rare detrimental variants can fluctuate greatly among oocytes. During oogenesis, mtDNA experiences a genetic bottleneck that only allows a small number of mtDNA to be amplified into several hundred thousand copies of mtDNA in mature oocytes. This could result in large swings in heteroplasmy levels between mother and child or between offspring (1,2,39). It is conceivable that some rare detrimental variants can resurface (due to genetic drift, mtDNA bottleneck, selection, or a combination of any of these processes) as high-level heteroplasmic variants in human pedigrees and contribute to the development of complex diseases during aging (3,40).

Further analysis showed that deleterious SNVs were prevalent among heteroplasmic SNVs in both human and mouse oocytes. A sizable portion of low-level heteroplasmic SNVs have been reported to be associated with cancers or mitochondrial diseases. Phylogenetic trees of individual mtDNA suggest a sequential accumulation of de novo mutations in the same mtDNA molecule. These observations raise interesting questions about the roles of these rare mtDNA mutations that exist at the beginning of one's life, and their implications for the diagnosis and prevention of mitochondrial diseases. Note that we only surveyed one of the most conservative lists of mitochondrial disease variants and many other mtDNA variants have been reported to link to mitochondrial diseases and complex diseases (3,26), so the overlap with disease variants may be underestimated. Although this study does not address the functional significance of these deleterious variants, it will be of great interest to perform longitudinal studies in the future to understand if and how they contribute to late-onset cancers or inherited mitochondrial diseases through selection and/or genetic drift (1,16,41,42). Our work demonstrated the possibility to follow the dynamics of such SNVs during development and aging in hope of deciphering the emergence of mutations and understanding their clinical significance.

Large mtDNA deletions are found in cells of mitochondrial disorder patients (43) and in aging postmitotic tissues (44,45). mtDNAs with large SVs are expected to have severe functional defects and are thought to be rarely transmitted in the germline (3). Heteroplasmic mtDNA deletions are postulated to occur sporadically soon after the mtDNA bottleneck in early development. The large number of mtDNA in oocytes are distributed among daughter cells in the early cell divisions of the embryo without

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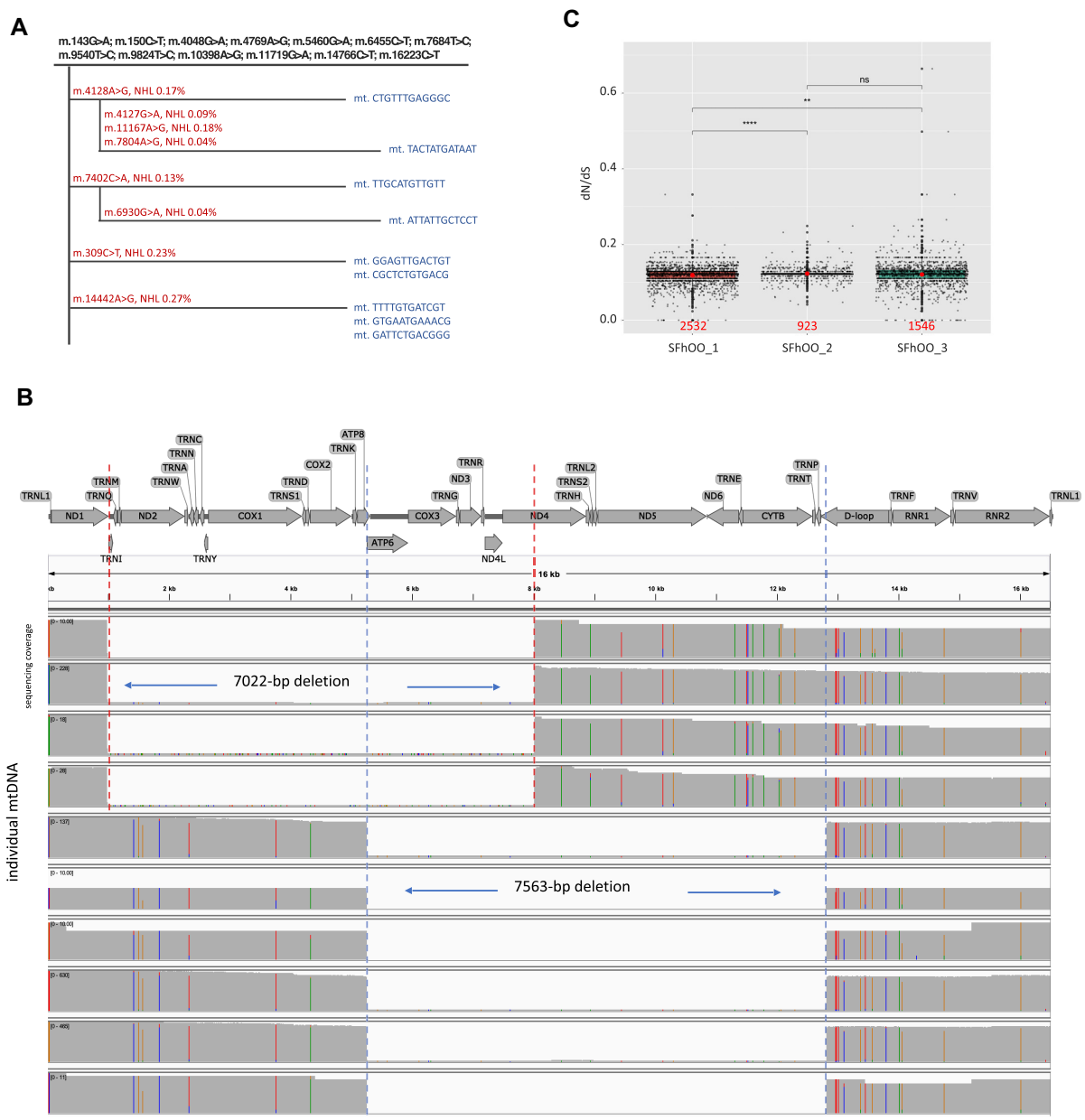


Figure 3. Genetic linkage analysis of mtDNA SNVs and SVs in single human oocytes. (**A**) Phylogenetic trees of individual detected mtDNA (shown in blue) constructed using ultra-rare SNVs (shown in Red) in the hOO_1 oocyte. Common SNVs (NHL > 1%) of the indicated mtDNA are shown in black. The DNA sequences indicate the UMIs. (**B**) A schematic map of the human mtDNA and the accompanying IGV tracks showing the alignment of Nanopore reads of ten UMI groups (mtDNA) harboring 7022-bp and 7563-bp deletions in SFhOO_2. Vertical dotted lines indicate deletion breakpoints. Colored bars in the read coverage track represent SNVs. (**C**) Box plots showing the distribution of dN/dS ratio of somatic SNVs (NHL < 60%) of individual mtDNA in SFhOOs. The mtDNA numbers of every sample are shown (red values). The red dots inside the boxes indicate the mean values of dN/dS, and the thick black lines represent the median values. Potential outlier values are marked by bold black dots, while individual values calculated from every UMI group are shown as small grey dots. The lower and upper boundaries of the box represent the 25th and 75th percentiles, respectively. The *P* values were based on the Kruskal-Wallis test, **** stands for $P < 2.22e\text{-}16$.

replication and deletion mutants can expand to high levels to affect multiple tissues (43). However, direct evidence of this has been lacking in the human germline. This is partly because short-read NGS of mtDNA necessitates fragmentation of mtDNA molecules, and it is unreliable in large SV analysis. Erroneous mapping of short reads to NUMTs can cause false variants (19). Hence, analysis of large SVs has hitherto been limited to Sanger sequencing of long PCR amplicons (44,45), which is low throughput and non-quantitative. Here, taking advantage of long-read sequencing and quantitative analysis of iMiGseq, we showed direct evidence of de novo large SVs in oocytes from a healthy donor. We also found surprising evidence of clonal expansion of SV-containing mtDNA, suggesting possible compensation by wild-type mtDNA in the nucleoid of the same mitochondria. Interestingly, the mtDNA with the 7022-bp

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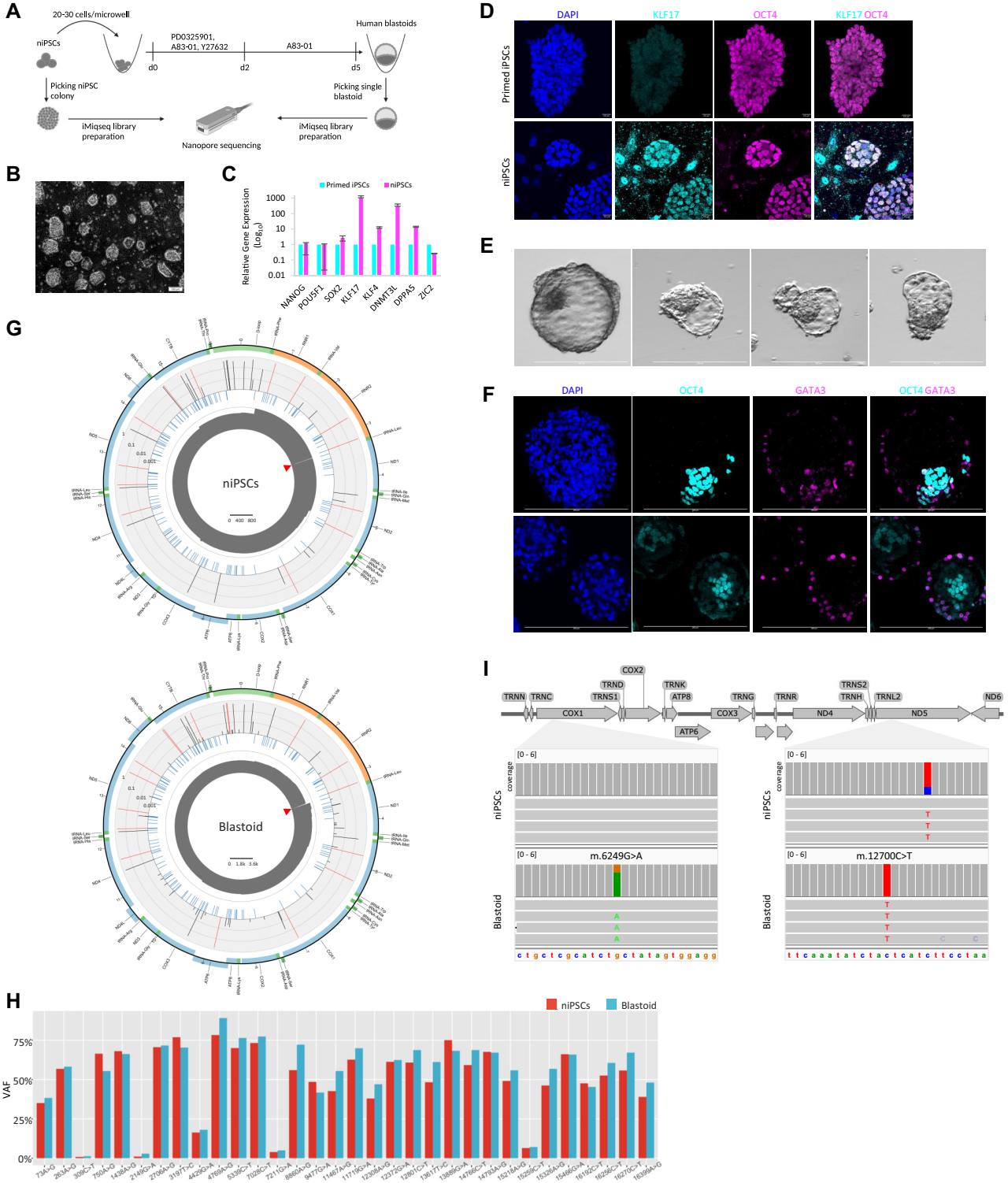


Figure 4. Tracking mtDNA heteroplasmy dynamics during the generation of human blastoid from niPSCs. (**A**) Schematics of human blastoid generation from niPSCs followed by iMiGseq. Created with Biorender.com. (**B**) Phase contrast image showing dome- shaped niPSC colonies that grew from single cells. Scale bar = 100 μm. (**C**) Quantitative real-time PCR data comparing primed iPSCs with niPSCs. NANOG, POU5F1 and SOX2 represent the core pluripotency genes. KLF17, KLF4, DNMT3L and DPPA5 represent naïve pluripotency markers. ZIC2 represents a primed pluripotency marker. (**D**) Immunofluorescence staining comparing primed iPSCs to niPSCs. KLF17 is a naïve marker, and OCT4 is a core pluripotency marker. Scale bar = 20 μm. (**E**) Representative widefield images for day 5 niPSC-derived blastoids. Scale bar = 200 μm. (**F**) Immunofluorescences staining of niPSC-derived day-5 blastoids with the ICM marker OCT4 and the trophectoderm marker GATA3. Scale bar = 200 μm. **(G)**, Circular plots showing the distribution of SNVs in the mitochondrial genome of niPSCs and single blastoid. The arrangement of the circular plots is similar to Figure 1A. (**H**) Comparison of heteroplasmy levels of primary SNVs (VAF > 1%). No significant VAF shift is observed. (**I**), Two rare amino-acid-changing SNVs of m.6249G > A (p.Ala116Thr) and m.12700C > T (p.Leu122Phe) detected in two individual mtDNA in the single blastoid but absent in niPSCs.

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deletion lacks the origin of light strand replication (Figure 3B), raising the question of how it is replicated. The mechanisms of mtDNA replication are still not fully understood. There is a controversy between two mtDNA replication models–a strand-displacement model and a strand-coupled model. Evidence supporting the strand-displacement model has been presented to demonstrate the need of the light strand origin in mtDNA replication. However, multiple light strand origins were purported by a previous study (46). In the strand-coupled model the light strand origin is not required. There have been reports suggesting that under certain conditions, strand-coupled replication may function as a backup replication mode in mammalian mitochondria (47,48). The molecular mechanisms underlying this type of replication have not been elucidated. Thus, it seems possible that the deletion mtDNA replicates itself using alternative light strand origins or through a strand-coupled replication mechanism.

Because iMiGseq enables, for the first time, quantitative base-resolution analysis of thousands of mtDNA in single cells, it may facilitate preimplantation genetic diagnosis (PGD) of mitochondrial diseases. For example, in the *in vitro* fertilization (IVF) setting, iMiGseq could be used for an unbiased analysis of all types of mtDNA mutations (SNVs and SVs) in single biopsied blastomeres, which have been shown to faithfully represent the heteroplasmy level of IVF embryos (49–52). Blastomere biopsy does not affect the development of the embryo and allows selection of embryos with the lowest mutation load for implantation. Current clinical protocols only screen for known mitochondrial disease variants and may miss novel variants or large SVs. Additionally, Sanger sequencing-based screening methods (53) are low-throughput and therefore limit the number of tests possible for low-input samples (i.e. a few blastomeres). The ability to unbiasedly discover mtDNA mutations in a few cells using iMiGseq could potentially overcome these challenges and improve the diagnosis of mitochondrial disease. Our proof-of-concept data in single human blastoid showed that this is not only feasible but also likely to yield novel knowledge of the full spectrum of mutations and their genetic linkage. To this end, human blastoids offer a promising platform to model and improve the clinical practice of PGD of mitochondrial disease. Besides the germline, it is logical to extend this study to somatic tissues to unravel the direction of causality between mtDNA mutations and aging and complex diseases in the future.

DATA AVAILABILITY

VAULT and sample data in this study are accessible at GitHub (https://github.com/milesjor/vault) and Zenodo (https://doi.org/10.5281/zenodo.7450392). Raw sequencing data are available in the SRA database (accession ID: PRJNA657296), which are accessible with the following link: https://www.ncbi.nlm.nih.gov/bioproject/PRJNA657296.

SUPPLEMENTARY DATA

Supplementary Data are available at NAR Online.

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Author contributions: C.B. and L.W. performed majority of the experiments related to sequencing. Y.F. and P.Z. performed the human oocyte collection and experiments related to hEPS cells. C.B. performed the bioinformatics analysis. B.Y. performed cell culture and mtDNA labeling experiments. S.M. performed cell culture and characterization related to niPSCs and blastoids. B.Y. and Y.Z. performed iMiGseq of blastoid experiments. C.B., J.C.I.B., Y.Y., Y.H. and M.L. analyzed the data. C.B. and M.L. wrote the manuscript. Y.H., Y.Y. and J.C.I.B. contributed to the writing of the manuscript. C.B. and M.L. conceived the study. Y.Y., J.C.I.B. and M.L. supervised the study.

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Conflict of interest statement. A patent application based on methods described in this paper has been filed by King Abdullah University of Science and Technology, in which CB, LW and ML are listed as inventors. The authors declare no other competing interest.

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